Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Quantum Biophotonic Integrated Circuits for the Early Detection of Sepsis: QBICs**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum (√)			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio & Medicine			
	Medium Category	Small Category	Medium Category	Small Category
	e.g. Semiconductor manufacturing process	e.g. Display manufacturing process		
Concept of Proposed Technology	• Semiconductor integrated circuit (IC) fabrication processing for biochips presents promising opportunities, including miniaturization, large-scale parallel processing, and the high-precision, cost-effective mass production of Lab-on-a-Chip devices. • Quantum biophotonic PCR IC are advanced optofluidic chips that utilize small semiconductor crystals to detect biological molecules, cells, and pathogens with exceptional sensitivity and accuracy. • Quantum biophotonic PCR IC for sepsis (QBICs) can demonstrate a promising future for high-resolution, multiplexed bioimaging and targeted drug delivery. This innovative technology offers substantial potential for advancing precision medicine.			
Rationale for Joint R&D Project	To enhance the molecular diagnostic biosensing platform for precision sepsis detection, it is essential to collaborate with companies capable of providing (1) highly precise semiconductor fabrication that enables nanometer-scale optical and electrical sensors for highly sensitive, label-free biomarker detection; and (2) scalable, cost-effective manufacturing by optimizing the utilization of existing large-scale semiconductor manufacturing facilities to reduce costs and improve access to advanced biochips.			

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Key Words	1 Semiconductor processing based nanopatterned electrodes 2. Nanoscale biophotonic sensor arrays 3. Multiplexed biophotonic PCR platform to improve accuracy and reliability 4. Massive fabrication process 5. Sample to answer point-of-care system

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)
R&D Objectives	**Quantum Biophotonic Integrated Circuits for the Early Detection of Sepsis: QBICs**			
R&D Target Specifications	• Semiconductor fabrication process-based multiplexed integrated optofluidic circuits. • Speed: reduced processing time (<3 min) for EVs purification compared to benchmark technologies (~8 hrs). • Higher exosome purity (>95%) versus benchmark technologies (~33%) for precise AD biomarker diagnosis • Sensitivity: enhanced sensitivity (< 1 copy/μl for nucleic acids) compared to benchmark technologies (~1 copy/μl) • Multiplexed sensing capabilities (>20 multiple pathogen biomarkers). • Product format: Standalone molecular diagnostic chip and AI system for sepsis. • Reliability: • Standardization: dependable molecular diagnostic tool for sepsis. • Acceleration of early detection of sepsis, providing an accessible screening option and enabling research groups to understand and prepare preventive medicine.			
R&D Plan	**Aim 1. Develop quantum biophotonic PCR chip capable of detecting multiplexed biomarkers (0 month – 18 months).** 1.1: Design and development of effective biophotonic PCR • Design and fabricate quantum biophotonic PCR chip. • Characterize quantum biophotonic PCR chip. • Evaluate batch-to-batch variation and reproducibility. 1.2: Develop mass-producible precise nanoplasmonic optical antennas • Develop and optimize semiconductor processes to obtain well-defined nanoplasmonic optical antennas on a wafer larger than 6”. • Characterize obtained nanoplasmonic optical antennas for optoelectronic properties. • Characterize the batch variations of nanoplasmonic optical antennas for the reproducibility of nanoplasmonic optical antennas fabrication. 1.3: Develop immobilization protocols of molecular probes on the nanoplasmonic optical antennas and rapid detection protocols • Study to define self-assembly monolayer and linkers to increase the number of molecular probes and improve alignment of nanoproves and stable immobilization on the nanopattern • Design and develop a rapid detection system using CMOS camera. • Optimize buffer to avoid non-specific biomolecules immobilization of molecular probes. 1.4: Develop highly sensitive and selective nanoplasmonic optical antennas-based quantum biophotonic PCR chip capable of detecting multiplexed biomarkers for the early detection of sepsis. • Assemble self-assembly monolayer, molecular probes, and active labeling molecules on the nanoplasmonic optical antennas and optimize reaction time and enzyme process. • Evaluate biosensing performance with biomarkers and analyze data for LOD, sensitivity, selectivity, and reproducibility. **Aim 2. Develop QBET as mobile multiplexed POC diagnostic platform (18 months – 28 months)** 2.1: Establish mobile and multiplexed POC diagnostic platform • Develop a QBET PCR on chip system using the well-defined nanoplasmonic optical antennas and characterize its detection range, LOD, sensitivity, selectivity, and reproducibility • Develop integrated nanofluidic sample preparation on chip and nanoplasmonic optical			

	antenna arrays to achieve multiplexed biomarker detection on a single chip and one-time detection • Evaluate PCR performance of integrated system with targeted biomarkers, analyze data for LOD, sensitivity, selectivity, and reproducibility, and then compare the obtained results with the bench top PCR system. 2.2: Validation of prototype integrated PCR system in the laboratory using biological samples • Establish a prototype of multiplexed QBET with nanofluidics POC kit • Validation of biomarkers in serum or blood and evaluate PCR chip performance with bench top PCR system. **Aim 3. Pre-clinical validation (29 months – 36 months)** 3.1: Validate cross-reactivity with different disease and reproducibility • Validate cross-reactivity of the QBET PCR chip as POC platform using samples containing multiple biomarkers • Compare with traditional blood culture method of detecting sepsis (3-6 hours) with multiplexed QBET PCR chip in terms of sensitivity, accuracy, and precision since PCR can detect pathogens 3.2: Pre-clinical performance assessment under various environmental conditions

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Development of a portable and high-performance multimodal POC platform for the early detection of sepsis** • Design quantum biophotonic PCR chip using nanoplasmonic optical antenna arrays and nanofluidic chip. • Develop immobilization protocols of nanoprobes on the nanoplasmonic optical antennas. • Develop highly sensitive and selective nanoplasmonic optical antenna -based biophotonic sensors capable of detecting multiplexed nucleic acids (DNAs and RNAs) biomarkers for the early disease detection of sepsis. • Integrating micro- or nanofluidic chip for sample preparation and reaction chambers and quantum nanoplasnmonic arrays for efficient and ultrafast PCR on chip. • Validation of integrated QBICS as a prototype of POC system in the laboratory using biological samples as pre-clinical testing data collection, and analysis • Disease-specific assays optimized for multiplexed detection • Real-world performance assessment under various environmental conditions **Intellectual Property (IP) & Technology Transfer** • Patent applications for immunoassay technologies, biosensor platforms, and integrated diagnostic devices • Licensing and commercialization strategies for industry adoption • Technology transfer to diagnostic companies for market deployment
Expected R&R of Korean SME Partner	**Develop and establish a protocol for mass-produced QBICs** **Develop QBICs using semiconductor IC manufacturing fabrication process.** • Design for manufacturability: define critical dimensions, pattern pitch/height, allowable tolerances, and wafer-level layout for high yield manufacring. • Process flow establishment: nanolithography → metal deposition → nanoplasmonic patterning → pattern and lyophilize PCR reagents • Thin-film deposition: sputtering/evaporation of Au/Pt/Ti/Cr layers; control of thickness for nanoplasmonic optical antennas. • Etch & surface engineering: RIE/wet etch optimization; surface activation (O_2 plasma/UV-ozone) for consistent surface chemistry for reliable • Yield & reproducibility study: DOE for key parameters (pressure, dose, temp/pressure, etch time); Cp/Cpk tracking across lots • Pattern and lyophilize PCR reagents help create enzyme-based master mixes that can be stored at room temperature. This innovation eliminates the need for cold-chain logistics, minimizes contamination risks, and makes rehydration simple, especially for point-of-care PCR chips.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Ultrafast multiplexed POC platform for precision medicine** • Established protocols of multiplexed NAs biochip. • Established semiconductor fabrication methods for nanofluidic biochips. • Established reproducible mass produced plasmonic circuits. • Established reproducible mass produced QDs on plasmonic circuits. • Integrated multiplexed molecular diagnostic biochip. • Validation of prototype POC kit in the laboratory using clinical samples. **Field evaluation and pre-clinical validation** • Real-world performance assessment under various environmental conditions **Cross-validated assay panels for osteoporosis disease** • Disease-specific assays optimized for multiplexed detection • Data on analytical performance, such as limit of detection, cross-reactivity, and reproducibility, is shared across sites for highly keen validation **Joint intellectual property and technology commercialization** • Technology licensing and commercialization planning through Korean SMEs and MGB • Filing of joint patents based on co-developed technologies **Strengthened Korea-MGB R&D Collaboration** • Established a long-term research network between academic, clinical, and industrial partners in both countries • Publication of joint peer-reviewed articles (SCI-indexed journal paper) and conference presentations to disseminate research findings **Commercialization potential** • **Product:** Portable POC system with recurring consumable sales (single-use cartridges-based biochips/reagents). • **Market fit:** Multiplexed monitoring of different diseases in decentralized settings (clinics, health-check centers, primary care). • **Scale-up:** Wafer-level nanoscale optical antennas on chip enable **mass production** and cost competitiveness via Korean SME/diagnostic partners. **Broader industrial/economic impact** • **Industry upgrading:** Strengthens the domestic value chain in **semiconductor biophotonic PCR chip + POC wireless AI system integration**. • **Healthcare impact:** Enables earlier/continuous monitoring to support better treatment decisions and reduce fracture-related burden. • **Economic spillover:** Creates high-value IP and commercialization pathways, supporting SME growth and export potential through Korea–MGB collaboration.